Advanced IO

File Mapping

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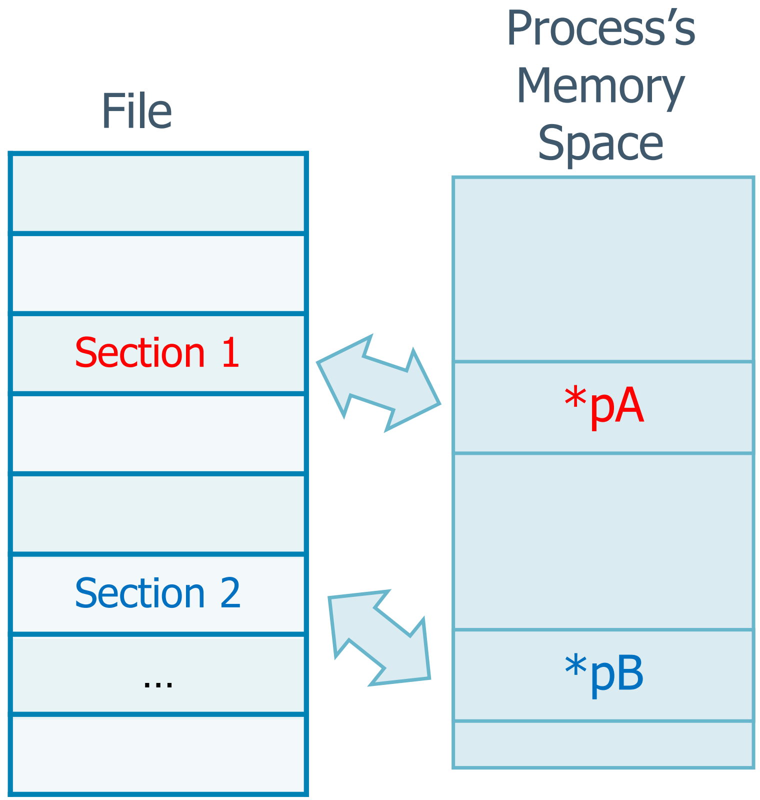
Memory Management

- OSs provide memory-mapped files to
  - Associate a file with the process's address space
  - Allow the OS to manage all data movement between the file and memory while the user just cope with memory address space
  - Permit the programmer to manipulate files **without** file IO functions
    - No need to use open, read, write, lseek, etc.



Memory Management

- The advantages to mapping normal files to the virtual memory space directly include
  - Applications can
    - Be significantly **faster**
      - **In-memory** algorithms (string processing, sorts, search trees) can directly process files
    - Manipulate larger quantity of data
      - Files may be much larger than the available physical memory
  - There is no need to manage buffers and the file data they contain
  - Multiple processes can **share** memory, and the file views will be coherent

Function mmap

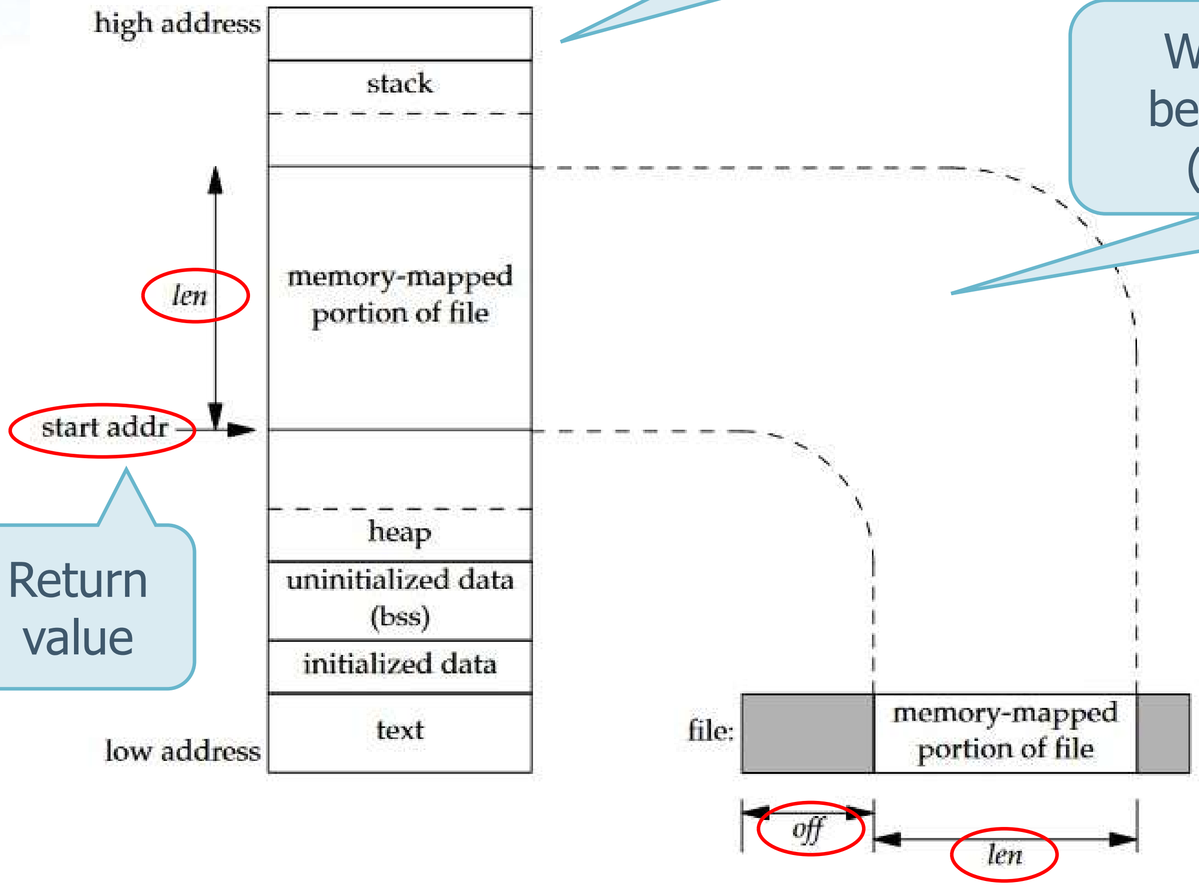
```
include <sys/mman.h>

void *mmap (
  void *addr,size_t len,int prot,int flag,int fd,off_t off
);
```

- We need to inform the the kernel to map a given file to a region in memory using mmap
- Return value
  - The starting address of the mapped region, in case of success
  - The constant MAP_FAILED, in case of errors

Function mmap

Layout of a typical process

We suppose the file is mapped between the heap and the stack (implementation dependent)



Return value

```
void *mmap (
  void *addr, size_t len, int prot,
  int flag, int fd, off_t off
);
```

Function mmap

- Parameters
  - **addr** specifies the address where we want the mapped region to start
    - We normally set this value to 0 to allow the system to choose the starting address
  - **len** is the number of bytes to map

```
void *mmap (
  void *addr, size_t len, int prot,
  int flag, int fd, off_t off
);
```

Function mmap

- **prot** specifies the protection of the mapped region
  - PROT_READ to read
  - PROT_WRITE to write
  - PROT_EXEC to execute
  - A bitwise OR of the previous flags
  - PROT_NONE indicates that the region cannot be accessed

```
void *mmap (
  void *addr, size_t len, int prot,
  int flag, int fd, off_t off
);
```

Function mmap

- **flag** affects various attributes of the region
  - MAP_FIXED
    - Forces the return value to be equal to addr
    - Its use is discouraged, as it hinders portability
  - MAP_SHARED
    - This flag specifies that a store operation is equivalent to a write to the file
    - Either this flag or the next (not both) must be specified
  - MAP_PRIVATE
    - Store operations into the mapped region cause a private copy of the mapped file to be created

```
void *mmap (
  void *addr, size_t len, int prot,
  int flag, int fd, off_t off
);
```

Function mmap

- **fd** argument is the file descriptor specifying the file that is to be mapped
  - We have to open this file before we can map it into the address space
- **off** is the starting offset in the file of the bytes to map

```
void *mmap (
  void *addr, size_t len, int prot,
  int flag, int fd, off_t off
);
```

Function munmap

```
include <sys/mman.h>

void *munmap (void *addr, size_t len);
```

- A memory-mapped region is unmapped when
  - The process terminates
  - We unmap a region by calling the **munmap** function
  - Note that closing the file descriptor does not unmap the region
- Return value
  - The value 0, in case of success
  - The value -1, in case of errors

Function munmap

- A call to munmap does not cause the contents of the mapped region to be written to the disk file
  - The kernel updates the disk file (for a MAP_SHARED region) automatically sometime after we write into the memory-mapped region
  - Modifications to memory in a MAP_PRIVATE region are discarded when the region is unmapped

Example

- Using memory mapping copy a source file into a destination (equivalente) file
  - In other words, implement the shell command “cp“ adopting memory mapped files
- The program receives the source and the destination file names on the command line

Example

We limit the amount of memory we copy to 1GByte to avoid exceeding the memory used

```
#include "apue.h"
#include <fcntl.h>
#include <sys/mman.h>

#define COPYINCR (1024*1024*1024)

int main(int argc, char *argv[]) {
  int fdin, fdout;
  void *src, *dst;
  size_t copysz;
  struct stat sbuf;
  off_t fsz = 0;

  fdin = open (argv[1], O_RDONLY));
  fdout = open(argv[2], O_RDWR|O_CREAT|O_TRUNC, FILE_MODE);
  if (fdin<0 || fdout < 0) error ...
  if (fstat(fdin, &sbuf) < 0) error ...
```

Example

See documentation for more details

```
#include <sys/types.h>
#include <sys/stat.h>

int stat (const char *path, struct stat *sb);
int lstat (const char *path, struct stat *sb);
int fstat (int fd, struct stat *sb);
```

The function stat returns a reference to the structure sb (struct stat) for the file (or file descriptor) passed as a parameter

```
struct stat {
  mode_t st_mode;/* file type & mode */
  ino_t st_ino;   /* i-node number */
  dev_t st_dev;   /* device number */
  dev_t st_rdev;  /* device number */
  ...
};
```

Some macros allow to understand the type of the file, e.g., S_ISREG(buf.st_mode) is true if the file is regular

The field **st_size** stores the size of the file

Example

```
while (fsz < sbuf.st_size) {
  if ((sbuf.st_size - fsz) > COPYINCR)
    copysz = COPYINCR;
  else
    copysz = sbuf.st_size - fsz;
  if ((src = mmap(0,copysz,PROT_READ,MAP_SHARED,fdin,fsz))
       == MAP_FAILED)
    ... error ...
  if ((dst = mmap(0, copysz, PROT_READ | PROT_WRITE,
       MAP_SHARED, fdout, fsz)) == MAP_FAILED)
    ... error ...
  memcpy(dst, src, copysz);
  munmap(src, copysz);
  munmap(dst, copysz);
  fsz += copysz;
  }
  return (0);
}
```

We iterates until the entire file is copied

We move on 1GByte at a time at most

Map source file

Map destination file

Copy

Unmap

Guidelines

- With mapping be careful with the file size
  - We cannot map too much
  - The size does not have to change
  - The mapped memory must be manipulated through pointers
    - Not need to use read and write system calls

```
...
memcpy(dst, src, copysz);
...
```

```
for (i=0; i<copysz; i++) {
  *dst = *src;
  dst++;
  src++;
}
```

Memory copy

Guidelines

- On modern UNIX-like operating system memory mapping can be faster or slower than direct read/write
  - With **read/write**, we activate more system calls
    - We copy the data from the kernel's buffer to the application's buffer (read), and then from the application's buffer to the kernel's buffer (write)
  - With **mmap/memcpy**, we copy the data directly from one kernel buffer mapped into our address space into another kernel buffer mapped into our address space